

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Robust diodes in SMC package

5.0SMDJxxS-HRA from Littelfuse are high-reliability TVS diodes that provide robust overvoltage protection with low early failures and zero degradation over continuous surge events. The diodes offer 60% higher surge handling capability and higher power rating than most other high-reliability TVS diodes. These come in a compact SMC package, offering higher power density, thus allowing more space-saving PCB designs. The diodes meet the DO-160 lightning protection regulatory requirement and offer a very low clamping voltage, making them suitable for a wide range of aircraft-related applications, including aircraft power trains and aircraft subsystems. They can also be used in harsh environments in applications such as AC or DC power line protection, low-frequency data lines, and industrial automation.

Littelfuse, Inc.

<https://www.littelfuse.com>

Integrated motor controller IC

ECS640A from Onsemi is a highly integrated motor controller that combines the controller and driver in a single product. The SiP is designed for reliable high-voltage operation of up to 600V. It offers a smaller form factor, thus reducing size, development cost, and the time to market. The controller is powered by an ARM



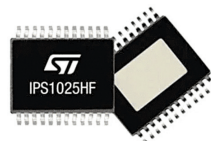
Cortex M0+ microcontroller and features three sense amplifiers and three bootstrap diodes. It can support both sensed and sensorless motor control architectures. The IC enables rapid reuse in alternative platforms by simply changing the discrete power devices and updating the software. It can simplify the development of high-voltage motor control systems in applications such as heating, venting and air-conditioning, refrigeration, and robotics.

Onsemi

<http://www.onsemi.com>

High-side power switch

IPS1025HF from STMicroelectronics is a fast-starting, high-side power switch for use in safety systems that require minimal power-on delay time. The switch has a response time of less than



60µs, making it suitable for critical applications. It can drive a resistive, inductive, or capacitive load with one side connected to the ground, thus simplifying the design and reducing component count. The compact device can work over a wide input voltage range of 8.65V to 60V and is able to tolerate up to 65V on the input pin. It offers a programmable current limitation value at power switch activation. IPS1025HF can enable electrical loads to instantly turn on and turn off by connecting it to a single side of the supply. It can withstand harsh industrial environments for applications such as vending machines, factory automation I/O peripherals, and com-

puter numerical control machines.

STMicroelectronics

<https://www.st.com>

Noise-canceling MEMS microphones

CUI Devices has announced a couple of new high-quality microelectromechanical systems (MEMS) microphones that are suitable for active



noise cancellation. The MM-2718AB-3815NC-TR and CMM-2718AT-3817NC-TR are analogue MEMS microphones that come in a compact package measuring just 2.75mm × 1.85mm × 0.90mm. Both the microphones are reflow solder compatible, which provides designers additional design flexibility as these components can be soldered at hard-to-reach corners. Both work over a frequency range of 20Hz to 20kHz and have a high signal-to-noise ratio (SNR). Their small size, high acoustic overload points, and high SNR makes them suitable for use in high-quality noise-cancelling microphone arrays.

CUI Devices

www.cuidevices.com

High-density STT-MRAM

EMxxLX family from Everspin Technologies is a high-density MRAM. The series offers the industry's first xSPI serial interface and is one of the highest-performing persistent memories available in the market. It offers a full read and write bandwidth of 400MB per second with a clock frequency of 200MHz. The STT-MRAM technology is less susceptible to the effects of radiation, which makes it highly reli-

